Everything Else

Cost management, front-end and mobile access

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AWS Budgets

- Create budget and **send alarms when costs exceeds the budget**
- 4 types of budgets: Usage, Cost, Reservation, Savings Plans
- For Reserved Instances (RI)
  - Track utilization
  - Supports EC2, ElastiCache, RDS, Redshift
- Up to 5 SNS notifications per budget
- Can filter by: Service, Linked Account, Tag, Purchase Option, Instance Type, Region, Availability Zone, API Operation, etc…
- Same options as AWS Cost Explorer!
- 2 budgets are free, then $0.02/day/budget

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Cost Explorer

- Visualize, understand, and manage your AWS costs and usage over time
- Create custom reports that analyze cost and usage data.
- Analyze your data at a high level: total costs and usage across all accounts
- Or Monthly, hourly, resource level granularity
- Choose an optimal **Savings Plan** (to lower prices on your bill)
- **Forecast usage up to 12 months based on previous usage**

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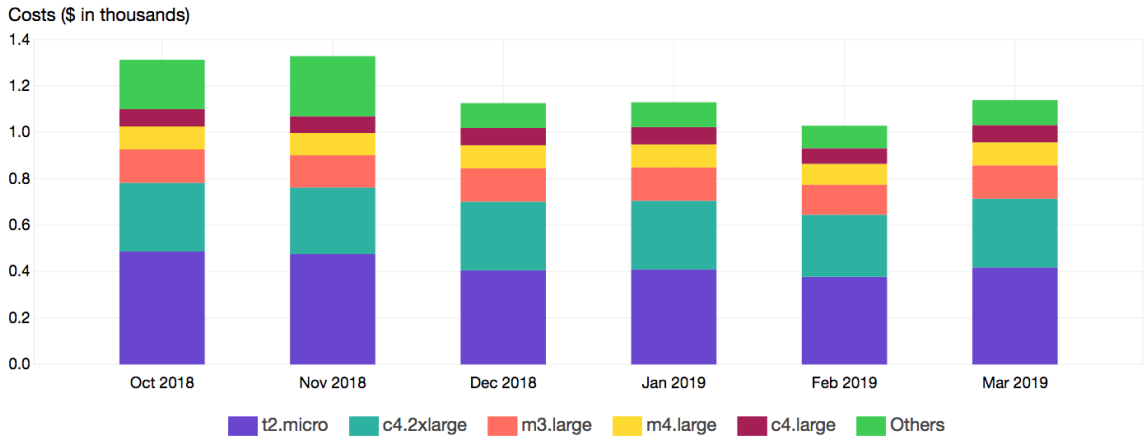
Cost Explorer – Monthly Cost by AWS Service

Monthly costs by service

Last 6 Months | Monthly | Stack

Group by: Instance Type × | Service | Linked Account | Region | Usage Type | Tag | API Operation | Availability Zone | More



Download CSV

Instance Type	Oct 1, 2018	Nov 1, 2018	Dec 1, 2018	Jan 1, 2019
Total cost ($)	1,312.71	1,328.54	1,125.99	1,129.65
t2.micro ($)	486.75	475.89	405.63	409.27
c4.2xlarge ($)	296.11	286.56	296.11	296.11

FILTERS | CLEAR ALL

Service — Include only
EC2-Instances × 1

Linked Account — Include all
Region — Include all
Instance Type — Include all
Usage Type — Include all
Usage Type Group — Include all
Tag — Include All
API Operation — Include all
Charge Type — Include all

More filters

ADVANCED OPTIONS

Show costs as
Unblended costs

Include costs related to
☐ Show only untagged resources

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Cost Explorer– Hourly & Resource Level

aws Services Resource Groups Admin/rosachae-Isengard @ 5... Global Support

AWS Cost Management > Cost Explorer: Cost & Usage Settings

Save as... Reports New report

Oct 25, 2019 - Oct 27, 2019 Hourly Stack

Group by: Resource Service Linked Account Region Instance Type Usage Type Tag More

Costs ($)

200 150 100 50 0

Oct-25* Oct-26* Oct-27*

i-0ab i-0ab i-0f33 i-06b9 Others

FILTERS CLEAR ALL

Service Include all

Linked Account Include all

Region Include all

Instance Type Include all

Usage Type Include all

Usage Type Group Include all

Resource Include all

Tag Include all

API Operation Include all

More filters

ADVANCED OPTIONS

Show costs as

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Cost Explorer – Savings Plan Alternative to Reserved Instances

Recommendation options

Savings Plans type	Savings Plans term	Payment option	Based on the past
◉ Compute	○ 1-year	◉ All upfront	○ 7 days
○ EC2 Instance	◉ 3-year	○ Partial upfront	○ 30 days
		○ No upfront	◉ 60 days

Recommendation: Purchase a Compute Savings Plan at a commitment of $2.40/hour

You could save an estimated $1,173 monthly by purchasing the recommended Compute Savings Plan.

Based on your past **60 days** of usage, we recommend purchasing a Savings Plan with a commitment of **$2.40/hour** for a **3-year term**. With this commitment, we project that you could save an average of **$1.61/hour** - representing a **40%** savings compared to On-Demand. To account for variable usage patterns, this recommendation maximizes your savings by leaving an average **$0.04/hour** of On-Demand spend.

Before recommended purchase	After recommended purchase (based on your past 60 days of usage)	
Monthly On-Demand spend	Estimated monthly spend	Estimated monthly savings
$2,955 ($4.05/hour)	**$1,782** ($2.44/hour)	**$1,173** ($1.61/hour)
Based on your On-Demand spend over the past 60 days	Your recommended $2.40/hour Savings Plans commitment + an average $0.04/hour of On-Demand spend	40% monthly savings over On-Demand $2,955 - $1,782 = $1,173

This recommendation examines your usage over the past 60 days (including your existing Savings Plans and EC2 Reserved Instances) and calculates what your costs would have been had you purchased the recommended Savings Plans. See applicable rates for Savings Plans here. To generate this recommendation, AWS simulates your bill for different commitment amounts and recommends the commitment amount that provides the greatest estimated savings. Learn more

Recommended Compute Savings Plans — Download CSV — Add selected Savings Plan(s) to cart

×	Term	Payment option	Recommended commitment	Estimated hourly savings
☑	3-year	All upfront	$2.40/hour	$1.61 (40%)

*Average hourly spend and minimum hourly spend based on your current on-demand spend for the given instance family.

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Cost Explorer – Forecast Usage

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Example: Building a Serverless API

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AWS API Gateway

- AWS Lambda + API Gateway: No infrastructure to manage
- Support for the WebSocket Protocol
- Handle API versioning (v1, v2…)
- Handle different environments (dev, test, prod…)
- Handle security (Authentication and Authorization)
- Create API keys, handle request throttling
- Swagger / Open API import to quickly define APIs
- Transform and validate requests and responses
- Generate SDK and API specifications
- Cache API responses

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API Gateway – Integrations High Level

- **Lambda Function**
  - Invoke Lambda function
  - Easy way to expose REST API backed by AWS Lambda
- **HTTP**
  - Expose HTTP endpoints in the backend
  - Example: internal HTTP API on premise, Application Load Balancer…
  - Why? Add rate limiting, caching, user authentications, API keys, etc…
- **AWS Service**
  - Expose any AWS API through the API Gateway
  - Example: start an AWS Step Function workflow, post a message to SQS
  - Why? Add authentication, deploy publicly, rate control…

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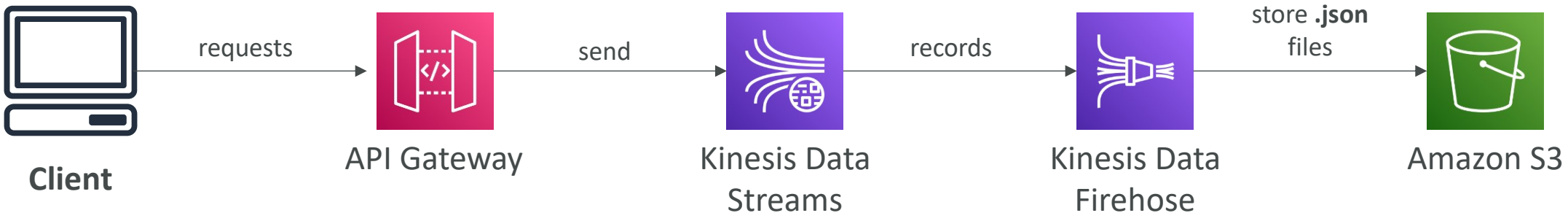
API Gateway – AWS Service Integration Kinesis Data Streams example



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API Gateway - Endpoint Types

- **Edge-Optimized (default):** For global clients
  - Requests are routed through the CloudFront Edge locations (improves latency)
  - The API Gateway still lives in only one region
- **Regional:**
  - For clients within the same region
  - Could manually combine with CloudFront (more control over the caching strategies and the distribution)
- **Private:**
  - Can only be accessed from your VPC using an interface VPC endpoint (ENI)
  - Use a resource policy to define access

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API Gateway – Security

- **User Authentication through**
  - IAM Roles (useful for internal applications)
  - Cognito (identity for external users – example mobile users)
  - Custom Authorizer (your own logic)

- **Custom Domain Name HTTPS** security through integration with AWS Certificate Manager (ACM)
  - If using Edge-Optimized endpoint, then the certificate must be in **us-east-1**
  - If using Regional endpoint, the certificate must be in the API Gateway region
  - Must setup CNAME or A-alias record in Route 53

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